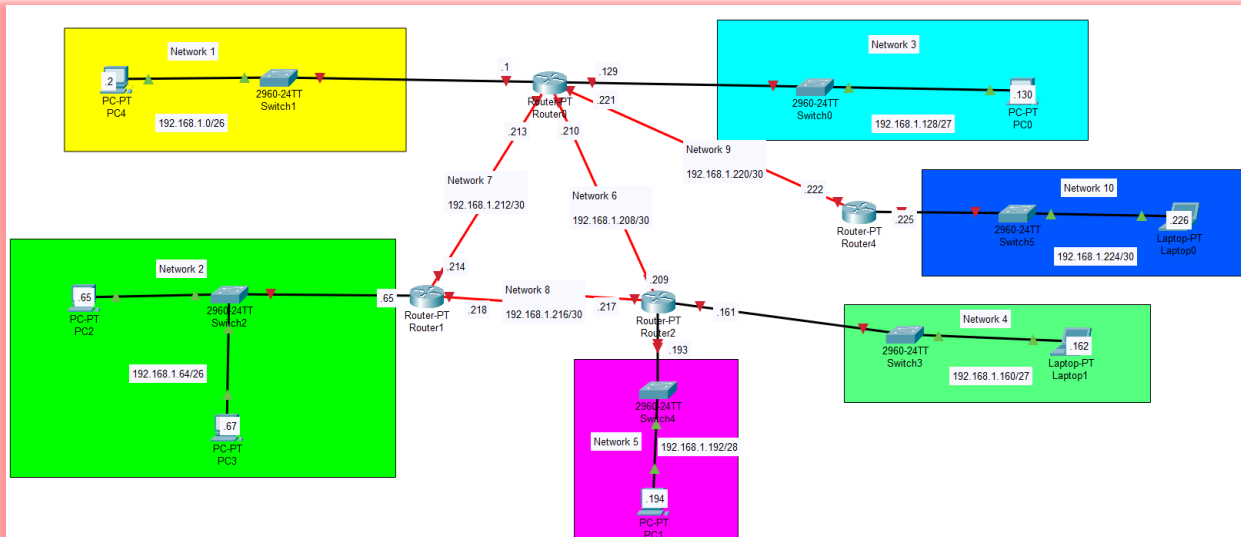


Lab 11 Tasks Report Submission

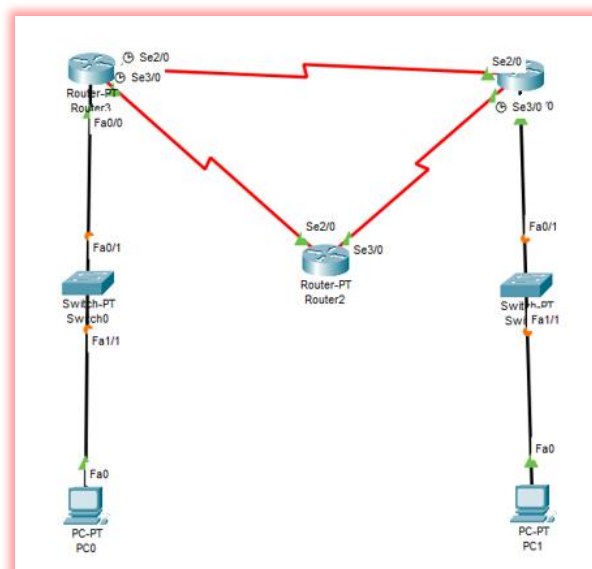
Task-01

Using Cisco Packet Tracer, simulate the network topology illustrated below, which consists of **4 major networks** and **10 subnetworks**. Assign appropriate IP addresses to all hosts and configure IP addresses for each router interface. Additionally, implement both **static routing** and a **default route** to ensure full connectivity across the network.



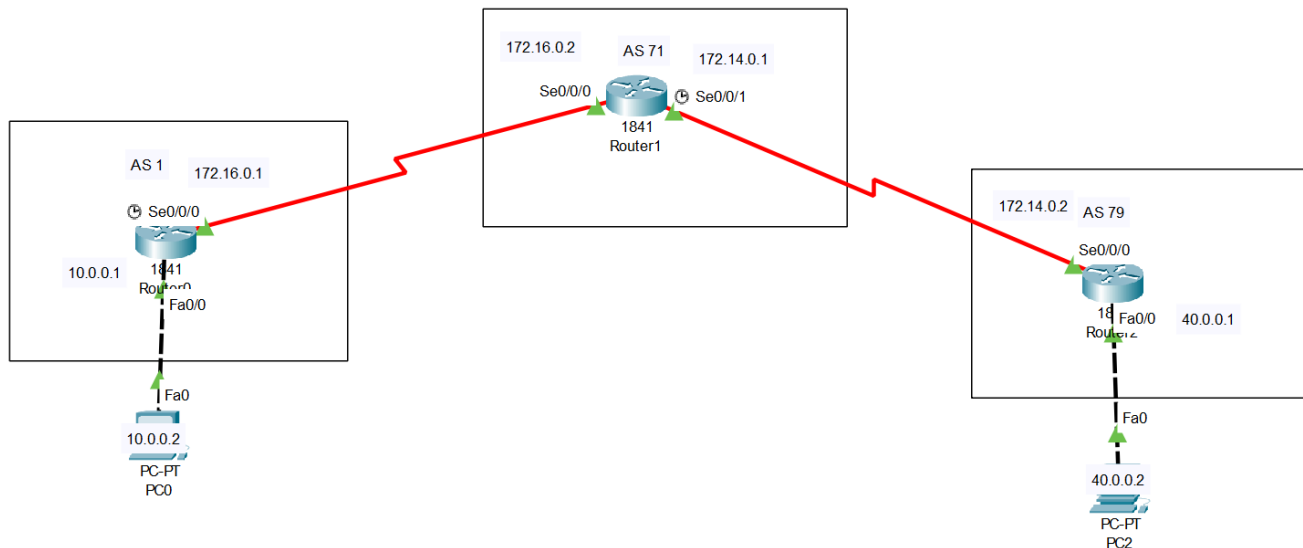
Task-02

Using Cisco Packet Tracer, simulate the given network topology consisting of multiple interconnected networks. Assign appropriate IP addresses to all devices and configure IP addresses for each network interface. Implement the OSPF (Open Shortest Path First) dynamic routing protocol to ensure efficient route advertisement and full connectivity across the network. Verify that all networks are properly advertised and reachable through OSPF.



Task-03

Using the network topology provided below, configure the Border Gateway Protocol (BGP) to enable inter-network communication between autonomous systems. Assign appropriate IP addresses to all interfaces and ensure that BGP is correctly implemented with accurate AS (Autonomous System) numbers. Verify proper route advertisement and successful connectivity between all participating networks.



Task-04

Nadeem, a senior network engineer, has been assigned the task of setting up Border Gateway Protocol (BGP) to connect two branch offices located in different autonomous systems (ASes). He collaborates with Sameer and Abdullah, who are managing the branch networks in AS 65001 and AS 65002, respectively. The main objective is to establish a reliable inter-domain routing setup that allows both branch offices to exchange route information securely and efficiently over the internet using BGP.

Each office has its own internal network, and a separate router connects each AS to a shared transit network representing the internet. The routers are pre-configured with basic IP addressing and are now ready for BGP implementation.

Tasks:

1. Configure BGP on each router with appropriate AS numbers (e.g., AS 65001 for Sameer, AS 65002 for Abdullah). Ensure that each router is aware of its BGP neighbors using the neighbor command.
2. Configure each router to advertise its internal LAN networks into BGP using the network command. Ensure that only the relevant networks are advertised, avoiding unnecessary or private address leakage.



3. Verify that the BGP peering is successfully established between the routers in different ASes. Use appropriate commands such as `show ip bgp summary` to confirm that the BGP neighbor relationship is active.
4. Perform end-to-end ping tests between PCs located in each AS to ensure full connectivity. If routes are not being received or advertised correctly, identify and resolve the misconfigurations. Additionally, document the steps and output used for verification.